Technical Document #30: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Content delivery networks (CDNs) distribute content globally to reduce latency.
- Multi-cloud strategies use services from multiple cloud providers.
- Containerization packages applications with their dependencies.